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# Navigating open-source software: Can you rely on what you use?

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# Key Principles of Science

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- Reproducible
- Replicable
- Trustworthy

> *Among the very basic principles that guide scientists, as well as many other scholars, are those expressed as respect for the integrity of knowledge, collegiality, honesty, objectivity, and openness.*

Responsible Science: Ensuring the Integrity of the Research Process: Volume I. 1992

# Trustworthy



- 16,554 packages available on CRAN
- <280,000 repositories using R on GitHub (most aren't packages though)

How do you find the "right" one(s) to use, the ones that you can trust?

> *I am a beginner R user. Since I've started to learn, I wonder who is it proved whether these packages work right? Who is checking for functions, particularly complicated ones whether they are written right? What is the best way of understanding a package is good or bad?*

Bahar Patlar's comment on Revolution Analytics 10,000 CRAN packages blog post.

# Sources of Trust

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- Recommendations from people you trust
- Reputation of authors
- Download / citation statistics
- Dependencies
- History of development
- Associated publications
- Listed on a CRAN Task View
- . . .

# Formalizing Trust

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- Validate CRAN packages
  - Unrealistic too large / expensive (money and time)
  - Can pay a company to validate for mission-critical packages
- Compare two independently developed packages in simple cases
- Create benchmarks for statistical software development
  - Not all “good” packages will adhere
  - What should these be?
  - How can these be both generic and specific?
- Create a risk strategy for R packages
  - What should the risk profile look like?
  - What attributes are less/more risky?

# rOpenSci






*rOpenSci fosters a culture that values open and reproducible research using shared data and reusable software*

- Package review - focused on “helper” packages
- Community calls
- Events
- Forum
- Statistical Software Peer Review - new project

# Statistical Software Peer Review

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Sponsored by 


- 6 board members (including me) and around 1.5 full-time staff
- Goal: develop a set of agreed-upon standards for statistical package implementation and testing
- Launch a new peer-review process

# Statistical Software Peer Review



It is really difficult!

- Statistics is diverse . . .
- . . . topic categories are diverse
- Thinking about general standards is tough

# Categories considered

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- Bayesian & Monte Carlo
- Dimensionality & Feature Reduction
- Machine Learning
- Regression, Splines, & Interpolation
- Statistical Indices and Scores
- Visualisation
- Exploratory Data Analysis (EDA)
- Probability Distributions
- Wrapper Packages
- Categorical Variables
- Networks
- Survival Analysis
- Workflow Software
- Summary Statistics
- Spatial Analysis
- Educational Software

# So far . . .

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- Currently looking at design and testing NOT correctness
- Have draft standards for 6 areas so far
  - Bayesian and Monte Carlo
  - Regression and supervised learning
  - Dimensionality Reduction, Clustering, and Unsupervised Learning
  - Exploratory Data Analysis
  - Time Series Software
  - Machine Learning Software
- First packages are undergoing review
- Seeking community review of the draft standards and welcome suggestions:

https://ropenscilabs.github.io/statistical-software-review-book/standards.html

# R Validation Hub

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*R Validation Hub is a cross-industry initiative whose mission is to enable the use of R by the Bio-Pharmaceutical Industry in a regulatory setting, where the output may be used in submissions to regulatory agencies.*

- Advocating risk assessment of packages
- Developing `riskmetric` R package
- Opportunity to contribute metrics as issues

https://github.com/pharmaR/riskmetric

# Conclusion

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- Selecting packages to use is often based on trust
- Trust in packages is hard to assess
- rOpenSci and R Validation Hub are trying to provide tools for users and developers

With prominent news stories about nationally important inference based on research code it is imperative that we address open-source software standards immediately.